

A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

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I. COPPER-PLATE STORAGE-SCHEDULING MODEL

A. Optimization Problem

A single storage device and a single grid connection serve a known demand p_L^t over T periods of length Δt , all quantities in per unit. Following the sign convention of [1], $P_B^t > 0$ discharges and $P_B^t < 0$ charges; B^t is the energy at the *start* of period t , so B^1 is given and B^{T+1} is terminal. With $t \in \mathcal{T} = \{1, \dots, T\}$:

$$\min_{P_{\text{Subs}}, P_B, B} \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t + w (B^{T+1} - B^1)^2 \quad (1a)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b)$$

$$B^1 = B_0 \quad (1c)$$

$$B^{t+1} = B^t - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d)$$

$$P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e)$$

$$P_B^{\min} \leq P_B^t \leq P_B^{\max}, \quad t \in \mathcal{T} \quad (1f)$$

$$B^{\min} \leq B^{t+1} \leq B^{\max}, \quad t \in \mathcal{T} \quad (1g)$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. Optimal-Control Form

Restated over $N = T$ stages in the form required by DDP, with a distinct cost and constraint at each stage:

$$\begin{aligned} \min_{x, u} \quad & \sum_{t=1}^{N-1} \ell^t(x^t, u^t) + \ell^F(x^N, u^N) \\ \text{s.t. } \quad & x^1 = \hat{x}^1, \\ & x^{t+1} = f(x^t, u^t), \quad t \in \{1, \dots, N-1\}, \\ & c^t(x^t, u^t) = 0, \quad t \in \{1, \dots, N\}. \end{aligned} \quad (2)$$

The state is the stored energy alone; the controls are the two power set-points:

$$x^t = B^t \in \mathbb{R}, \quad u^t = (P_{\text{Subs}}^t, P_B^t)^\top \in \mathbb{R}^2, \quad (3)$$

with $\hat{x}^1 = B_0$. Because stage t carries its own c^t and p_L^t , the per-period data enters directly and nothing is interpolated:

$$f(x, u) = x - u_2 \Delta t, \quad (4)$$

$$c^t(x, u) = u_1 + u_2 - p_L^t, \quad (5)$$

$$\ell^t(x, u) = [c^t u_1 + C_B u_2^2] \Delta t, \quad (6)$$

$$\ell^F(x, u) = [c^T u_1 + C_B u_2^2] \Delta t + w (x - u_2 \Delta t - B_0)^2. \quad (7)$$

Bounds (1e)–(1f) map onto control limits $b^L \leq u^t \leq b^U$. The state bound (1g) is appended as a second row of c^t via a slack s^t adjoined to the control, giving $n_c = 2$, $n_u = 3$:

$$(x - u_2 \Delta t) - B^{\min} - s^t = 0, \quad 0 \leq s^t \leq B^{\max} - B^{\min}. \quad (8)$$

Remark. A per-stage ℓ^t and c^t is what the method assumes — the objective, dynamics and constraint functions are time-varying in general — but the reference implementation exposed only one of each for the whole horizon. An earlier revision of this benchmark worked around that by adjoining a time index τ to the state and recovering c^t and p_L^t from a degree- $(T-1)$ Lagrange interpolant of it. That device is exact at the nodes and was numerically well behaved past $T = 24$, but its symbolic expression grows with the horizon and exhausted the code generator at $T = 48$. Restoring per-stage data removes it, and with it the extra state: n_x falls from 2 to 1 and the horizon ceiling disappears.

C. What the Solver Actually Forms

Section I-B states the model in the solver's generic objects x, u, ℓ, f, c . This subsection eliminates them: every quantity below is written in the variables of (1) and the multipliers attached to its constraints. All expressions are verified against the derivative functions the solver actually builds.

1) Variables carried at stage t

The state is the stored energy, and nothing else:

$$\underbrace{B^t}_{\text{state}}; \quad \underbrace{P_{\text{Subs}}^t, P_B^t, s^t}_{\text{controls}}. \quad (9)$$

Each stage carries one multiplier per constraint:

$$\varphi_{\text{bal}}^t : P_{\text{Subs}}^t + P_B^t - p_L^t = 0 \quad (10)$$

$$\varphi_{\text{soc}}^t : (B^t - P_B^t \Delta t) - B^{\min} - s^t = 0 \quad (11)$$

$$\nu_{\text{Subs}}^t : P_{\text{Subs}}^t \geq 0 \quad (12)$$

$$\nu_{\text{min}}^t, \nu_{\text{max}}^t : P_B^{\min} \leq P_B^t \leq P_B^{\max} \quad (13)$$

$$\lambda_{B \text{ min}}^t, \lambda_{B \text{ max}}^t : 0 \leq s^t \leq B^{\max} - B^{\min}. \quad (14)$$

The last line is the correspondence that matters: because s^t is defined by (11) as the slack of B^{t+1} above $B^{\min}$, its two box multipliers *are* the energy-bound multipliers, $\lambda_{B \text{ min}}^t$ on $B^{t+1} \geq B^{\min}$ and $\lambda_{B \text{ max}}^t$ on $B^{t+1} \leq B^{\max}$. There is no separate state-constraint dual. P_{Subs}^t has no upper bound, so no multiplier is attached to one.

2) Stage derivatives

With the dynamics $B^{t+1} = B^t - P_B^t \Delta t$ linear and the constraints (10)–(11) linear in the controls, every second derivative of f and of c^t vanishes, as does the cross term

$\partial^2 \ell^t / \partial u \partial B$. What remains is

$$\frac{\partial c}{\partial (P_{\text{Subs}}, P_B, s)} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -\Delta t & -1 \end{bmatrix} \quad (15)$$

$$\frac{\partial c}{\partial B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (16)$$

$$\frac{\partial B^{t+1}}{\partial P_B^t} = -\Delta t \quad (17)$$

$$\frac{\partial^2 \ell}{\partial (P_{\text{Subs}}, P_B, s)^2} = \text{diag}(0, 2C_B \Delta t, 0). \quad (18)$$

The terminal stage adds $2w$ to the B curvature, $2w(\Delta t)^2$ to the P_B curvature, and $-2w\Delta t$ to their cross term.

3) The matrix that is factorised

Write V_{BB}^{t+1} for the curvature of the cost-to-go in the stored energy. Collecting (17)–(18) with the barrier terms, the stage Hessian is *diagonal*:

$$\hat{H}^t = \text{diag} \left(\underbrace{\frac{\nu_{\text{Subs}}^t}{P_{\text{Subs}}^t}}_{P_{\text{Subs}}}, \underbrace{2C_B \Delta t + (\Delta t)^2 V_{BB}^{t+1} + \Sigma_{P_B}^t}_{P_B}, \underbrace{\Sigma_s^t}_s \right), \quad (19)$$

$$\Sigma_{P_B}^t = \frac{\nu_{\min}^t}{P_B^t - P_B^{\min}} + \frac{\nu_{\max}^t}{P_B^{\max} - P_B^t}, \quad (20)$$

$$\Sigma_s^t = \frac{\lambda_{\min}^t}{s^t} + \frac{\lambda_{\max}^t}{B^{\max} - B^{\min} - s^t}. \quad (21)$$

Three controls and two equalities leave a *one*-dimensional null space, so after eliminating (15) exactly one direction survives:

$$d = (\Delta P_{\text{Subs}}, \Delta P_B, \Delta s) = (-1, +1, -\Delta t), \quad (22)$$

which is the statement that the only freedom left at a stage is to move power between the substation and the battery, the energy slack following. The quantity the solver factorises is therefore the *scalar*

$$d^\top \hat{H}^t d = \frac{\nu_{\text{Subs}}^t}{P_{\text{Subs}}^t} + 2C_B \Delta t + (\Delta t)^2 V_{BB}^{t+1} + \Sigma_{P_B}^t + (\Delta t)^2 \Sigma_s^t, \quad (23)$$

and the Cholesky whose failure triggers regularisation is a positivity test on this one number. Dropping the energy bound removes s^t and its two multipliers, leaving $d = (-1, +1)$ and $d^\top \hat{H}^t d = \nu_{\text{Subs}}^t / P_{\text{Subs}}^t + 2C_B \Delta t + (\Delta t)^2 V_{BB}^{t+1} + \Sigma_{P_B}^t$; the base instance strips the barrier terms too, leaving $2C_B \Delta t + (\Delta t)^2 V_{BB}^{t+1}$, which is where C_B enters as the curvature that keeps the problem solvable.

4) What is passed backward

Let V_B^t denote the gradient of the cost-to-go in the stored energy. With a one-dimensional state, V_B^t and V_{BB}^t are scalars, so the whole backward recursion is scalar arithmetic. The backward recursion carried between stages is

$$V_B^t = V_B^{t+1} + \varphi_{\text{soc}}^t + (\beta^\top \hat{Q}_u^t)_B + (\omega^\top c^t)_B, \quad (24)$$

$$V_{BB}^t = V_{BB}^{t+1} + (\beta^\top \hat{B}^t)_{BB} + (\omega^\top c_x^t)_{BB}, \quad (25)$$

where β and ω are the feedback gains produced by the stage solve. At a feasible iterate ($c^t = 0$) that has converged ($\hat{Q}_u^t = 0$), (24) collapses to

$$V_B^t - V_B^{t+1} = \varphi_{\text{soc}}^t, \quad (26)$$

TABLE I
FILTERDDP ITERATIONS, ALL BOUNDS

k	objective	backward pass				forward pass	
		pr_∞	du_∞	cs_∞	$\log_{10} \mu$	α	bk
0	0.840000	1.0e+00	9.2e-01	1.0e+00	-1.40	0.50	1
1	1.225783	5.0e-01	4.6e-01	3.8e-01	-1.40	0.50	1
2	1.397002	2.5e-01	2.3e-01	1.9e-01	-2.10	0.50	1
3	1.464625	1.2e-01	1.1e-01	9.6e-02	-2.10	1.00	0
4	1.520935	2.2e-16	7.8e-16	1.8e-02	-2.80	0.50	1
5	1.507179	2.6e-16	6.1e-16	1.1e-02	-3.49	0.50	1
6	1.496237	2.5e-16	1.2e-15	6.0e-03	-3.49	0.50	1
7	1.488496	2.2e-16	7.8e-16	3.5e-03	-4.19	0.50	1
8	1.483187	3.3e-16	1.1e-15	2.0e-03	-4.19	0.50	1
9	1.479927	3.3e-16	1.3e-15	1.1e-03	-4.19	1.00	0
10	1.476349	2.2e-16	7.5e-16	3.1e-04	-5.03	1.00	0
11	1.475395	2.2e-16	1.8e-15	7.0e-05	-6.04	1.00	0
12	1.475195	2.2e-16	3.5e-15	1.0e-05	-6.04	1.00	0
13	1.475179	4.4e-16	1.3e-14	1.4e-06	-7.25	1.00	0
14	1.475173	4.4e-16	1.9e-15	6.7e-08	-8.70	1.00	0
15	1.475173	2.2e-16	6.3e-15	2.0e-09	-10.44	1.00	0
16	1.475173	2.2e-16	1.1e-13	3.7e-11	-10.44	—	—

The backward pass evaluates the residuals at the current iterate; the forward pass is the filter line search that follows it, with accepted step α and bk = $\log_2(1/\alpha)$ backtracks. FilterDDP prints these two halves one iteration out of step, because `iteration_status` is called before `forward_pass!`; the rows here are realigned so each holds one backward pass and the forward pass it actually fed. The last row has no forward pass — the solver converged instead of stepping. Regularisation never fired. Every backtrack in the table was a rollout rejection rather than a filter or sufficient-decrease rejection, and the solver's own `ls` counter records none of them.

i.e. the value gradient in stored energy is the discrete costate, and its stage-to-stage increment is exactly the multiplier on the energy relation. Equation (25) has no counterpart in a first-order method: it is the curvature, and it is the object that a scheme passing only (26) between stages does not carry.

5) What is passed forward

With step size γ and $\delta B = B^t - \bar{B}^t$ the deviation from the nominal trajectory, every quantity is updated by the same affine rule:

$$P_{\text{Subs}}^t = \bar{P}_{\text{Subs}}^t + \gamma \alpha_{\text{Subs}}^t + \beta_{\text{Subs}}^t \delta B, \quad P_B^t = \bar{P}_B^t + \gamma \alpha_B^t + \beta_B^t \delta B, \quad (27)$$

$$\varphi_\bullet^t = \bar{\varphi}_\bullet^t + \gamma \psi_\bullet^t + \omega_\bullet^t \delta B, \quad \nu_\bullet^t = \bar{\nu}_\bullet^t + \gamma \chi_\bullet^t + \zeta_\bullet^t \delta B, \quad (28)$$

so the primal variables, the equality multipliers and the bound multipliers are all propagated together along the same search direction, with a single γ accepted by the filter for all of them.

D. Closed-Form Reference

The base instance is a strictly convex equality-constrained QP. Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to

$$J(P_B) = \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t (P_B^t)^2 + w(\Delta t)^2 s^2, \quad (29)$$

with $s = \sum_t P_B^t$. Stationarity divided through by Δt is $-c^k + 2C_B P_B^k + 2w\Delta t s = 0$; summing over k collapses it to

$$s = \frac{\sum_t c^t}{2C_B + 2Tw\Delta t}, \quad P_B^k = \frac{c^k - 2w\Delta t s}{2C_B}. \quad (30)$$

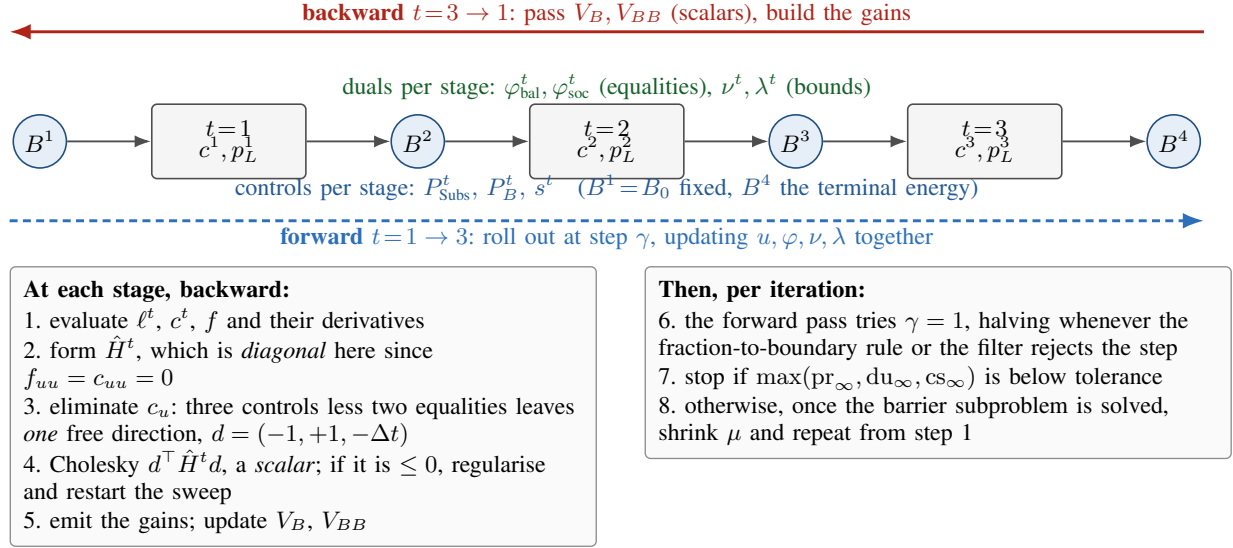


Fig. 1. How the copper-plate problem is actually solved, drawn at $T = 3$ so the tape fits; the structure is the same at any horizon, and the results reported here are $T = 6$. Read it as one worked example. Each stage owns its own price and demand, so nothing is interpolated and the state is the stored energy alone — which is why the cost-to-go derivatives V_B and V_{BB} carried backward along the top are single numbers rather than a vector and a matrix. The backward sweep builds a local model at each stage and leaves behind an affine rule; the forward sweep applies those rules left to right at one step size γ , moving the powers, the equality multipliers and the bound multipliers together. The stage solve is the interesting part: three controls against two equalities leave exactly one direction free, so the “dense” factorisation is a division by a single number, and the regularisation DDP is known for fires only if that number turns non-positive — which on this problem it never did.

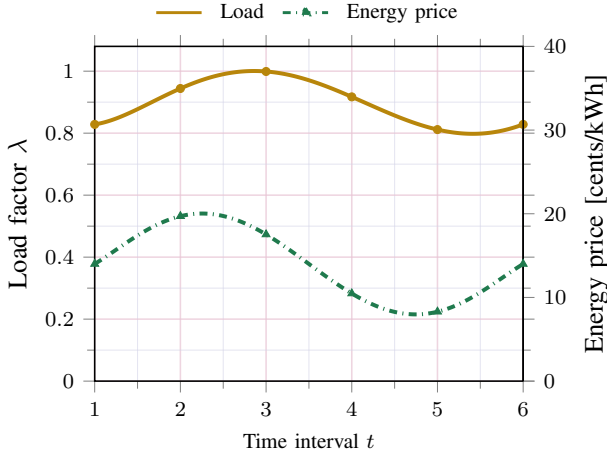


Fig. 2. Input profiles at $T = 6$: load factor (left axis), energy price (right). The markers are the data the solver uses — one price and one demand per stage, taken directly; the curves are the underlying tADMM profiles they are sampled from, drawn only to show the shape. The -0.8 rad phase offset between them is what keeps the two series from being collinear.

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, “Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks,” in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.

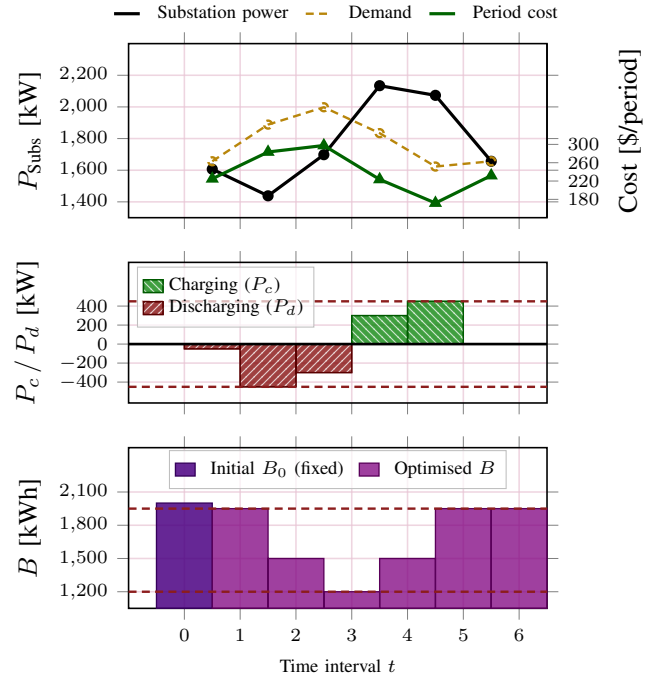


Fig. 3. The benchmark at $T = 6$ with all bounds imposed. Charging is drawn upward, discharging downward, so a bar’s direction is the device’s action rather than the sign of P_B^t . The battery discharges through $t = 1$ – 3 and recharges at $t = 4, 5$, riding its ± 450 kW rails and both energy limits (dashed). Those five bounds are the active set the reference solver reports.

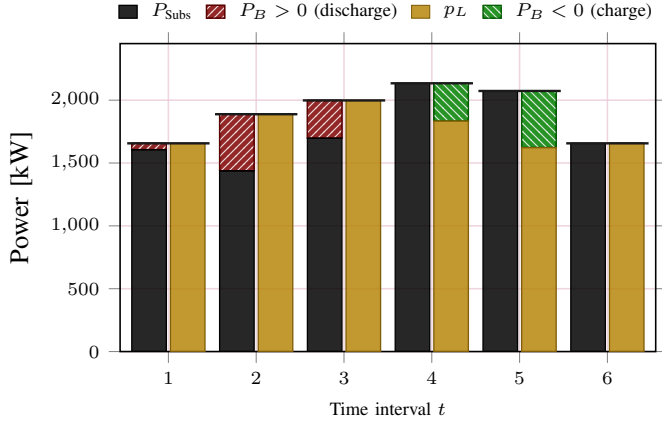


Fig. 4. Power-balance composition at $T = 6$. Per interval, supply (P_{Subs}^t plus discharge) on the left against demand (p_L^t plus charge) on the right; the equal heights are the power balance, marked by the rule. At $t = 2$ the battery supplies 23.8% of the node's 1888 kW; at $t = 5$ its charging instead raises import to 2073 kW to serve 1623 kW of load.

TABLE II
INSTANCE DATA

Interval t	1	2	3	4	5	6
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140

Period length Δt	1	Initial energy B_0	2.0
Throughput C_B	0.05	Terminal weight w	5.0

Bounds, constrained variants only

$P_B^{\min}, P_B^{\max}$	-0.45, 0.45
$B^{\min}, B^{\max}$	1.20, 1.95

All quantities are in per unit on the tADMM base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh; with $\Delta t = 1$, power and energy are numerically interchangeable, and the bounds are ± 450 kW and 1200–1950 kWh. Figs. 3 and 4 report battery quantities in kW and kWh rather than per unit. The series are the tADMM load and price profiles, $p_L^t = 2 \lambda^t$ with $\lambda^t = 0.8 + 0.1(\sin(\theta^t - 0.8) + 1)$ and $c^t = 0.08 + 0.06(\sin \theta^t + 1)$, sampled at $\theta^t \in \text{range}(0, 2\pi, T)$. These *resample* with T , so a shorter horizon is not a prefix of this row; both endpoints of $[0, 2\pi]$ are included, which is why $c^1 = c^T$. The -0.8 phase offset on the load is what keeps the two series from being collinear; their Pearson correlation is 0.644. Bounds apply only to the constrained variants.

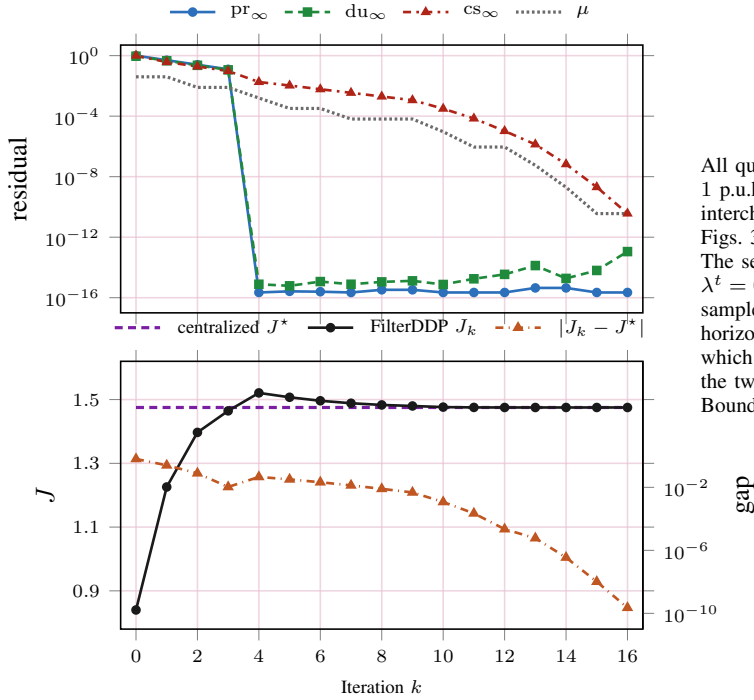


Fig. 5. FilterDDP convergence with all bounds imposed, 17 iterations. Top: the KKT residuals and the barrier parameter. Feasibility is reached almost immediately — pr_∞ and du_∞ hit machine precision by $k = 4$ — after which progress is entirely μ driving cs_∞ down alongside it, so the iteration count is set by the barrier schedule rather than by the optimisation. Bottom: the objective against the centralized JuMP/Ipopt baseline J^* , with the gap on the right axis. J_k is not monotone — it overshoots J^* near $k = 5$ and returns from above, because the line search decreases the barrier objective rather than J — and the gap closes to 2.3×10^{-10} .